

Quick Recall: Tree characterizations

Tree (Def. 5.1.1). A graph G is *acyclic* if it contains no cycle. A *tree* is a connected acyclic graph. A vertex of degree 1 is a *leaf*.

End-vertex Lemma (5.1.3). Every tree on ≥ 2 vertices has at least 2 leaves.

Tree-growing Lemma (5.1.4). If v is a leaf of G , then G is a tree iff $G - v$ is a tree.

Theorem 5.1.2. The following are equivalent for a graph $G = (V, E)$:

- (i) G is a tree.
- (ii) (*Path uniqueness*) Between every $x, y \in V$ there is exactly one path.
- (iii) G is connected, and removing any edge disconnects G .
- (iv) G is acyclic, and adding any non-edge creates a cycle.
- (v) G is connected and $|V| = |E| + 1$.

Practice: Textbook Exercise 5.1.3* — a path-uniqueness characterization

Let $n \geq 3$ and let $G = (V, E)$ be a graph on n vertices. Prove that the following are equivalent:

- (I) G is a tree.
- (II) $G \not\cong K_n$, and for every non-edge $\{x, y\} \in \binom{V}{2} \setminus E$ there is exactly one path in G from x to y .
 - (a) [Easier direction] Prove (I) $\Rightarrow$ (II). *Hint: Why is K_n not a tree for $n \geq 3$? After ruling that out, which condition in Theorem 5.1.2 does (II) follow from?*
 - (b) [Harder direction] Prove (II) $\Rightarrow$ (I).

Hints

For (b), break the proof into two steps.

Step 1: connected. Adjacent pairs use the edge; non-adjacent pairs use (II). (*One line.*)

Step 2: acyclic. *By contradiction.* Assume G has a cycle; let $C = v_0 v_1 \cdots v_{k-1} v_0$ be a *shortest* cycle.

- **Case $k \geq 4$.** First show C has no chord (a chord would split C into two strictly shorter cycles). Then $\{v_0, v_2\}$ is a non-edge, and the two arcs of C give two distinct paths from v_0 to v_2 .
- **Case $k = 3$.** The triangle's three pairs are all edges, so we must look *outside* it.

Key trick #1: extend the triangle to a maximal clique H . (Maximality replaces an induction: *any* $w \notin H$ is forced to have a non-neighbor in H .) Some $w \notin H$ exists (else $G = K_n$). Two scenarios share the same shape—in both, find an “entry vertex” $h_0 \in H$, then use the clique to detour:

- (A) w has a neighbor $h_0 \in H$. By maximality, w has a non-neighbor $h \in H$ ($h \neq h_0$). For any $h_1 \in H \setminus \{h_0, h\}$, two paths from w to h :

$$w \rightarrow h_0 \rightarrow h \text{ (length 2)} \quad \text{vs.} \quad w \rightarrow h_0 \rightarrow h_1 \rightarrow h \text{ (length 3)}.$$

- (B) w has *no* neighbor in H . Take a shortest path $Q: w \rightsquigarrow h_0 \in H$.

Key trick #2 — switch target. Pick *any* $h \in H \setminus \{h_0\}$; then $\{w, h\}$ is automatically a non-edge. For any $h_1 \in H \setminus \{h_0, h\}$, two paths from w to h :

$$w \rightsquigarrow h_0 \rightarrow h \text{ (length } m+1) \quad \text{vs.} \quad w \rightsquigarrow h_0 \rightarrow h_1 \rightarrow h \text{ (length } m+2).$$